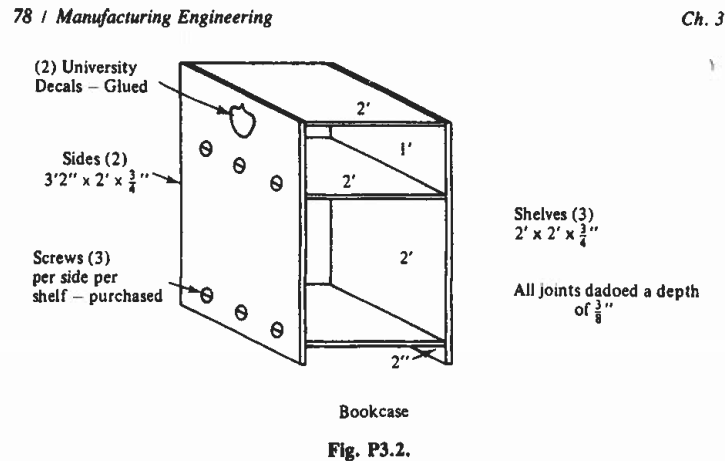


INDU 211/2X 2025 – Assignment #1

- Submit your answers on Moodle as one pdf file.

Question #1

Prepare a Bill-of-Material (BOM) for the following product (Chapter 3, Problem 2 from the textbook). The BOM should be presented with a “tree” structure, similar to that in the 3rd set of lecture slides on Manufacturing Engineering.



2. Prepare a bill of material for the product shown in Figure P3.2.

Question #2

Based on the discussions in class and in the textbook, for **ONE** of the processes below,

- assembling a wooden chair, you bought from IKEA in a package of a complete set of pieces
 - assembling a BBQ stove delivered in a box with all components
 - assembling a kitchen faucet from pieces you purchased from Home Depot.
 - or assembling any other consumer product to be assembled with components which you are familiar with (give clear introduction and explanations on the product and the assembling process)
- Briefly describe the main components of the product and the key activities required to assemble it. Prepare an Operations Process Chart (similar to the one in the lecture slides on Manufacturing Engineering) for the tasks.
 - Assuming that you will setup an “Assembly Line” to assemble large number of the same product. Briefly discuss:
 - how would you make proper arrangement of the “assembly line”?
 - the economic factors you need to consider for this “assembly production”.